Math 22 Course at a Glance

Geometry

Topical units are not numbered because instructional order may vary. Objectives are grouped by curriculum priority; supporting skills are intentionally omitted from this summary.

Review Content

REVIEW	
M22-REV-001	I can solve a linear equation arising from a geometric relationship.
M22-REV-002	I can solve a proportion and interpret the resulting scale factor.
M22-REV-003	I can simplify an exact square root and approximate it when needed.
M22-REV-004	I can calculate distance, midpoint, and slope on the coordinate plane.
M22-REV-005	I can substitute measurements into a formula with consistent units.

Geometry Basics

REQUIRED	
M22-BAS-001	I can name and represent points, lines, planes, segments, and rays using correct notation.
M22-BAS-002	I can identify collinear, coplanar, intersecting, and concurrent figures.
M22-BAS-003	I can calculate segment lengths using betweenness and the segment-addition postulate.
M22-BAS-004	I can determine a midpoint or use a segment bisector relationship.
M22-BAS-005	I can classify and measure angles.
M22-BAS-006	I can calculate angle measures using angle addition and angle bisectors.
M22-BAS-007	I can identify and use complementary, supplementary, vertical, and linear-pair relationships.
M22-BAS-008	I can distinguish a definition, postulate, theorem, converse, and counterexample.
M22-BAS-009	I can determine angle relationships formed by parallel lines and a transversal.
M22-BAS-010	I can use angle relationships to determine whether lines are parallel.
M22-BAS-011	I can solve problems involving parallel, perpendicular, vertical, or horizontal lines.
M22-BAS-012	I can construct a logical geometric argument from stated facts and accepted results.

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Triangle Congruence and Proof

REQUIRED	
M22-CON-001	I can classify triangles by sides and angles.
M22-CON-002	I can use triangle angle-sum and exterior-angle relationships.
M22-CON-003	I can write a congruence statement with corresponding vertices in the correct order.
M22-CON-004	I can determine whether triangles are congruent by SAS.
M22-CON-005	I can determine whether triangles are congruent by SSS.
M22-CON-006	I can determine whether right triangles are congruent by HL.
M22-CON-007	I can determine whether triangles are congruent by ASA or AAS.
M22-CON-008	I can apply isosceles and equilateral triangle theorems and their converses.
M22-CON-009	I can use corresponding parts of congruent triangles after proving congruence.
M22-CON-010	I can write a structured proof involving triangle congruence.

Quadrilaterals, Perimeter, and Area

REQUIRED	
M22-QUAD-001	I can calculate interior or exterior angle measures of a polygon.
M22-QUAD-002	I can calculate perimeter and area of common polygons.
M22-QUAD-003	I can apply properties of parallelograms to determine unknown measures.
M22-QUAD-004	I can determine whether a quadrilateral is a parallelogram.
M22-QUAD-005	I can apply properties of rectangles, rhombi, and squares.
M22-QUAD-006	I can classify a quadrilateral from its stated or marked properties.
M22-QUAD-007	I can apply trapezoid and trapezoid-midsegment relationships.
M22-QUAD-008	I can apply properties of kites.
M22-QUAD-009	I can calculate the perimeter or area of a composite polygon.
M22-QUAD-010	I can justify a quadrilateral conclusion using definitions and theorems.

Similarity and Proportional Reasoning

REQUIRED	
M22-SIM-001	I can determine image lengths and coordinates under a dilation.
M22-SIM-002	I can identify the scale factor of a dilation or similarity relationship.
M22-SIM-003	I can write a similarity statement with corresponding vertices in the correct order.
M22-SIM-004	I can use proportional corresponding sides to solve similar-polygon problems.
M22-SIM-005	I can relate similarity scale factor to perimeter and area scale factors.
M22-SIM-006	I can prove triangles similar by AA, SSS, or SAS.
M22-SIM-007	I can apply triangle proportionality theorems and their converses.
M22-SIM-008	I can solve and justify an indirect-measurement or other similarity application.

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Right Triangles

REQUIRED	
M22-RGT-001	I can apply the Pythagorean theorem to determine a missing side.
M22-RGT-002	I can use the converse of the Pythagorean theorem to classify a triangle.
M22-RGT-003	I can apply triangle-inequality relationships.
M22-RGT-004	I can determine exact side lengths in 45-45-90 and 30-60-90 triangles.
M22-RGT-005	I can apply right-triangle similarity relationships.
M22-RGT-006	I can use geometric-mean relationships created by an altitude to the hypotenuse.
M22-RGT-007	I can select sine, cosine, or tangent for a right-triangle problem.
M22-RGT-008	I can determine a missing side or angle of a right triangle.
M22-RGT-009	I can solve a contextual right-triangle problem.
M22-RGT-010	I can justify a right-triangle solution using an appropriate theorem or ratio.

Circles

REQUIRED	
M22-CIR-001	I can identify and name radii, diameters, chords, secants, tangents, arcs, sectors, and segments.
M22-CIR-002	I can calculate circumference and arc length.
M22-CIR-003	I can calculate the area of a circle, sector, or segment.
M22-CIR-004	I can determine angle and arc measures involving central and inscribed angles.
M22-CIR-005	I can apply relationships among chords, arcs, and distances from the center.
M22-CIR-006	I can apply tangent-radius and tangent-segment relationships.
M22-CIR-007	I can determine angle measures formed by chords, secants, or tangents.
M22-CIR-008	I can determine segment lengths formed by intersecting chords.
M22-CIR-009	I can determine segment lengths formed by secants or tangents.
M22-CIR-010	I can solve a multi-step circle problem and justify each relationship used.